SEMICONDUCTOR DEVICES TECHNOLOGY

Practical file

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Semester - VI

Course – B.Sc.(H) Electronics

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EXPERIMENT – 1

Aim: - To study the Czochralski (CZ) technique for single crystal growth and simulate the dopant distribution (segregation) using Scilab**.**

Apparatus Required: - Scilab Software, PC/Laptop

Theory: - The Czochralski (CZ) technique is the primary industrial method for producing high-quality single-crystal semiconductors like Silicon (Si).

The Process Physics:

1. **Melting:** High-purity polycrystalline material is melted in a quartz crucible.
2. **Seeding:** A perfectly oriented seed crystal is dipped into the melt.
3. **Pulling & Rotation:** The seed is slowly withdrawn and rotated. As the liquid atoms cool at the interface, they align with the seed's crystal lattice, forming a cylindrical ingot.

Dopant Segregation:

When a crystal grows from a melt containing impurities (dopants), the concentration of the dopant in the solid (C_s) is usually different from the concentration in the liquid (C_i). This is defined by the Equilibrium Segregation Coefficient (k_0):

$$k_0 = C_s / C_i$$

- If $k_0 < 1$: The impurity prefers the liquid; the crystal gets purer at the start, and the melt becomes more "polluted" as growth continues.
- If $k_0 > 1$: The impurity prefers the solid; the crystal takes in more do

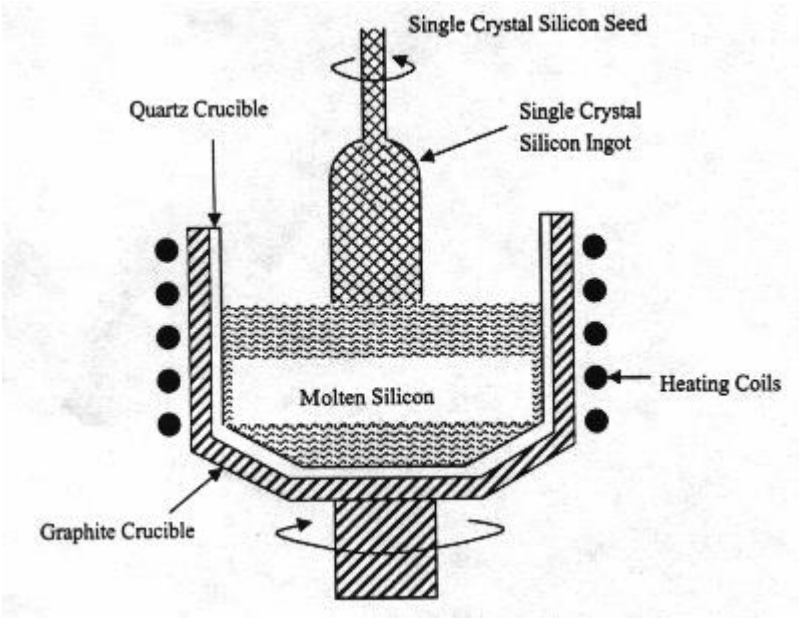
Formula Used: -

$$C_s / C_i = k_0(1 - f)^{(k_0 - 1)}$$

Where:

- C_s: Concentration of dopant in the solid crystal.
- C_i: Initial doping concentration in the melt.
- f: The fraction of the melt that has solidified (ranges from 0 to 1).
- k_0: Equilibrium segregation coefficient.

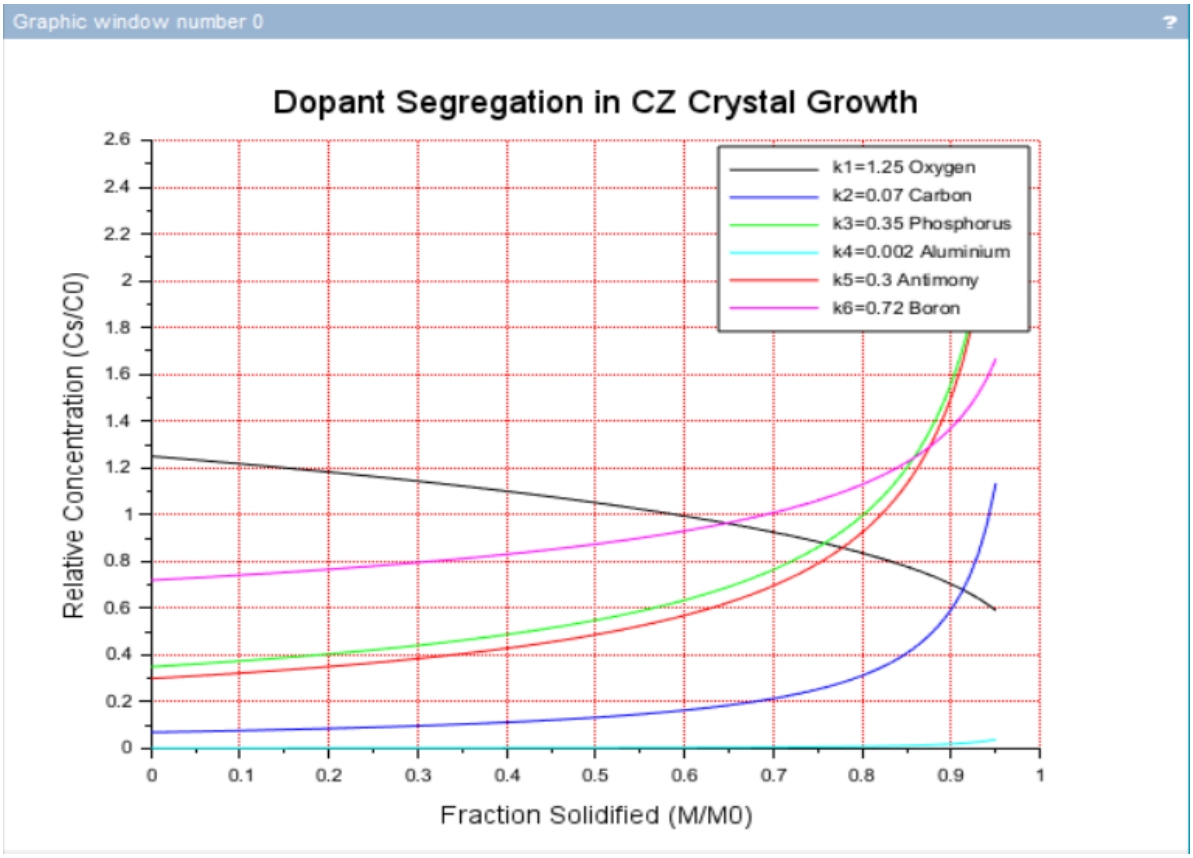
Diagram: -



Code: -

```
clf();
x = linspace(0, 0.95, 100);

k0 = [1.25, 0.07, 0.35, 0.002, 0.3, 0.72];
labels = ["k1=1.25 Oxygen", "k2=0.07 Carbon", "k3=0.35 Phosphorus", ...
          "k4=0.002 Aluminium", "k5=0.3 Antimony", "k6=0.72 Boron"];

for i = 1:6
    y = k0(i) * ((1 - x).^(k0(i) - 1));
    plot2d(x, y, style=i);
end

title("Dopant Segregation in CZ Crystal Growth", "fontsize", 4);
xlabel("Fraction Solidified (M/M0)", "fontsize", 3);
ylabel("Relative Concentration (Cs/C0)", "fontsize", 3);
legend(labels, "lower right");
xgrid(5);
```

Output: -



Observation: -

The generated plot illustrates how different dopants behave during the crystal pulling process:

1. **Dopants with $k_0 < 1$ (e.g., Aluminium, Phosphorus):** The curves rise sharply as M/M_0 approaches 1. This shows that these impurities are "pushed" into the remaining melt, making the tail-end of the crystal much more heavily doped than the seed-end.
2. **Dopants with $k_0 > 1$ (e.g., Oxygen):** The curve starts high and decreases. This indicates the dopant is depleted from the melt as the crystal grows.
3. **Conclusion:** The CZ technique requires precise control over the pulling rate to maintain a uniform dopant profile, as the chemical composition of the melt constantly changes during the process.

Application: -

- **VLSI Fabrication:** Creating the silicon wafers used for microprocessors.
- **Optoelectronics:** Growing crystals for LEDs and Laser diodes.
- **Renewable Energy:** Production of monocrystalline silicon for high-efficiency solar cells.

EXPERIMENT – 2

Aim: - To study the Float Zone (FZ) technique

Apparatus Required: - Scilab Software, Laptop

Theory: - The Float Zone (FZ) technique is a specialized crystal growth method used to produce ultra-high purity silicon (or other semiconductors). Unlike the Czochralski (CZ) method, the FZ method does not use a crucible. This is its biggest advantage, as it completely eliminates oxygen and carbon contamination that typically leaches from the quartz/graphite crucibles used in the CZ process.

The Physical Principle:

1. A polycrystalline silicon rod is held vertically.
2. A single crystal "seed" is attached to the bottom.
3. A radio frequency (RF) heating coil surrounds the rod. It creates a narrow, localized region of melted silicon (the "molten zone").
4. The RF coil slowly moves upward. As it moves, the molten zone travels up the rod. The polycrystalline material melts at the top boundary of the zone, while single-crystal silicon solidifies at the bottom boundary, adopting the structure of the seed crystal.

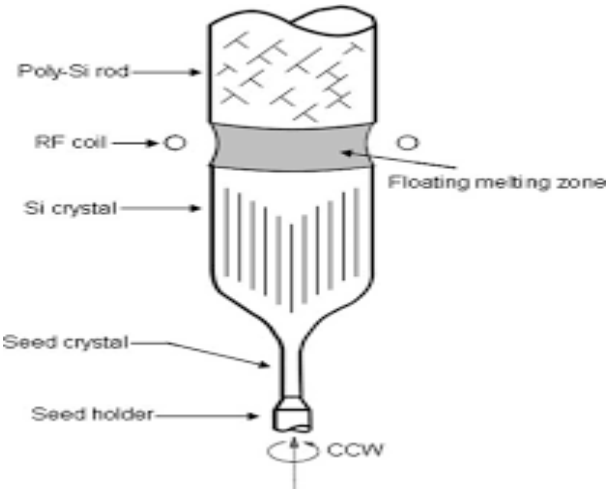
Formula Used: -

$$C_s(x) = C_0[1 - (1 - k_0)\, e^{-kx/L}]$$

Where:

- $C_s(x)$: Concentration of the dopant in the solid at position x
- C_0: Initial uniform doping concentration in the polycrystalline rod.
- k_0: Equilibrium segregation coefficient of the specific impurity.
- L: Length of the molten zone.
- x: Distance along the rod from the starting point

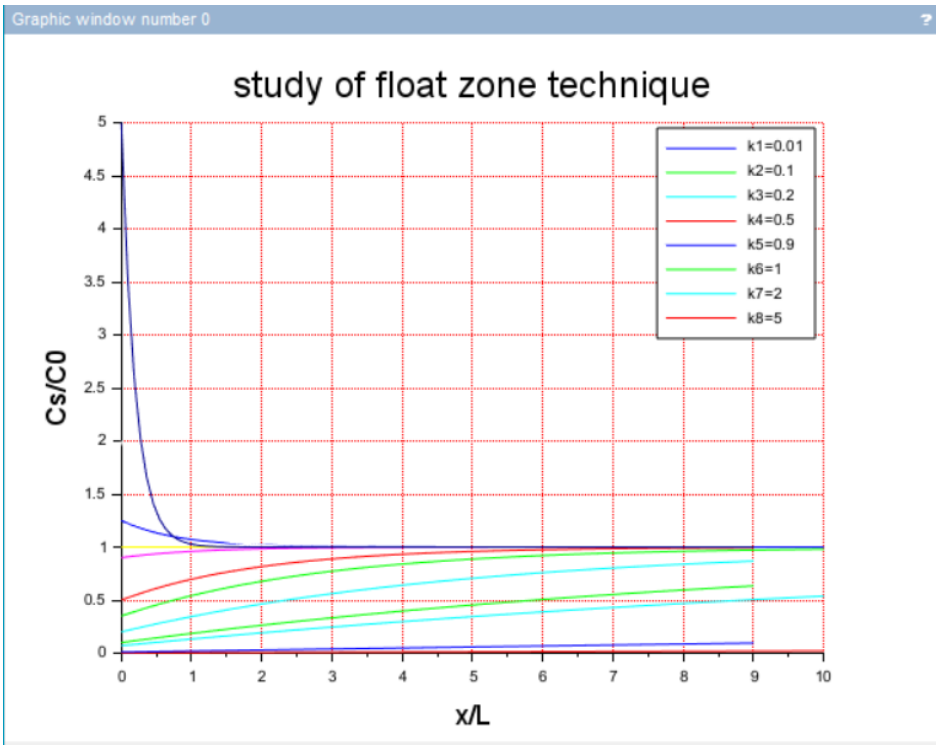
Diagram: -



Code: -

```
clc;
x=linspace(0,9,100);
ke=[0.01 0.1 0.2 0.5 0.9 1 2 5]
for i=1:1:8
y=1-(1-ke(i))*(exp((-ke(i)*x)));
plot2d(x,y,style=i+1)
end
title("study of float zone technique","fontsize",5);
xlabel('x/L',"fontsize",4);
ylabel('Cs/C0',"fontsize",4);
legend("k1=0.01","k2=0.1","k3=0.2","k4=0.5","k5=0.9","k6=1","k7=2","k8=5");
```

Output: -



<u>Observation</u>: -

By observing the generated graph, we can analyze how the Float Zone technique redistributes impurities:

- **For $k_0 < 1$:**
  The concentration starts low (equal to k_0) and slowly increases along the rod, finally reaching 1. This means impurities stay in the molten region, so the initial part of the crystal becomes purer. Lower $k_0$gives better purification.
- **For $k_0 = 1$:**
  The concentration remains constant at 1 throughout. There is no change because the impurity behaves the same in both solid and liquid.
- **For $k_0 > 1$:**
  The concentration is high at the beginning and then decreases towards 1. This shows impurities prefer the solid phase, so they accumulate at the starting end of the crystal.

<u>Application</u>: -

- **High-purity silicon production:** Used to make ultra-pure silicon wafers (very low oxygen content), essential for high-performance electronics.
- **Power electronics**: Widely used in devices like power diodes, thyristors, and IGBTs where high resistivity and purity are required.
- **Semiconductor research:** Ideal for studying impurity distribution and crystal defects because of its controlled growth process.

EXPERIMENT – 3

Aim: - To study constant total dopant concentration diffusion process

Apparatus Required: - Scilab Software, Laptop

Theory: - Diffusion is an important process in semiconductor fabrication used to introduce dopant atoms into silicon. In the constant total dopant concentration diffusion process (also called limited source diffusion), a fixed amount of dopant is deposited initially and no additional dopant is supplied during diffusion.

During heating, dopant atoms move from regions of high concentration to low concentration, following Fick's law. As time increases, the dopant spreads deeper into the semiconductor while the total amount of dopant remains constant. The concentration profile follows a Gaussian distribution.

The Physical Principle:

- A known quantity of dopant is initially deposited on the semiconductor surface.
- The wafer is heated to a high temperature.
- Dopant atoms diffuse from regions of high concentration to low concentration.
- Over time, the surface concentration decreases while the dopant penetrates deeper into the semiconductor.

Formula Used: -

$$C(x,t) = \frac{Q}{\sqrt{\pi D t}} e^{-x^2/4Dt}$$

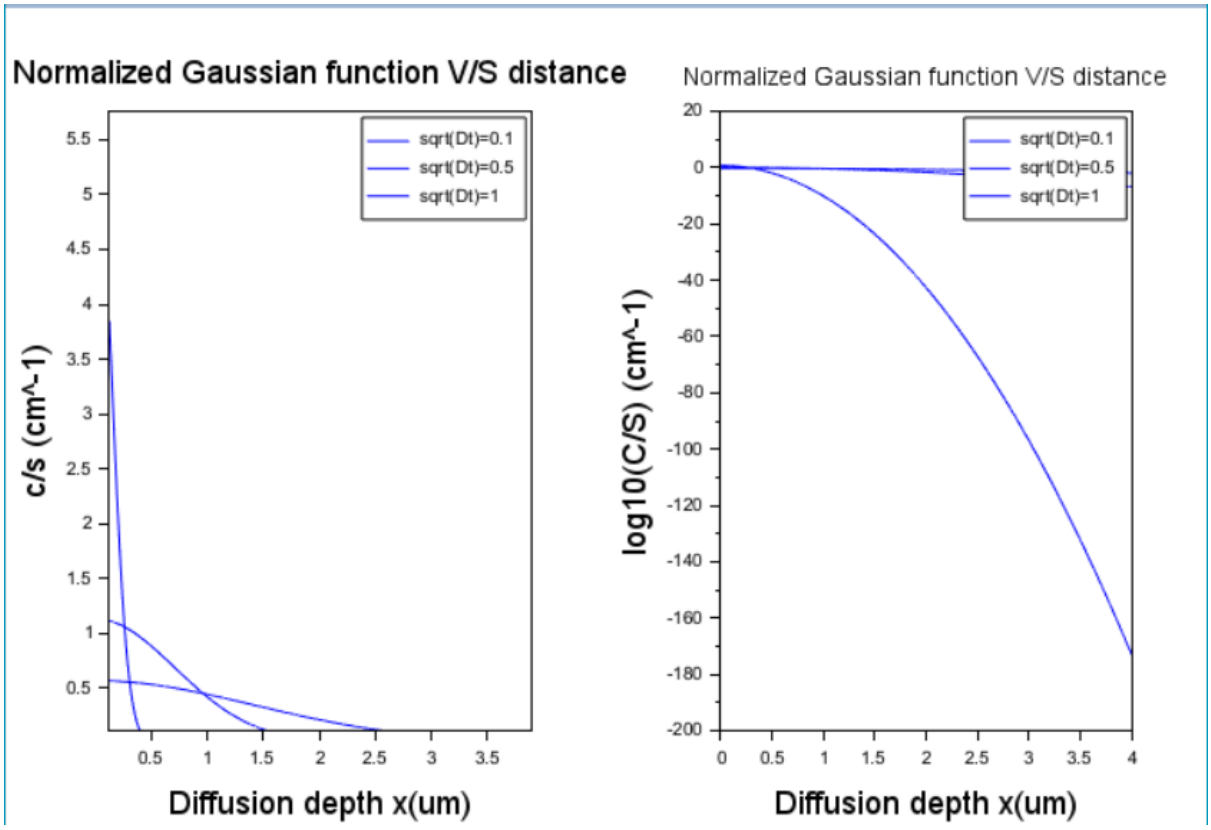
Where:

- $C(x,t)$: Dopant concentration at depth x
- Q: Total dopant amount
- D: Diffusion coefficient
- t: Diffusion time
- x: Depth from the surface

Code: -

```
Dt = [0.1 0.5 1]; // sqrt(Dt)
x = 0:0.001:4;

// Left plot (C/S vs x)
subplot(121);
for i = 1:3
    y = (1/(sqrt(%pi)*Dt(i))) * exp(-(x.^2)/(4*(Dt(i)^2)));
    plot(x, y, style=i, rect=[0 0 4 6]);
end

legend("sqrt(Dt)=0.1","sqrt(Dt)=0.5","sqrt(Dt)=1");
xlabel("Diffusion depth x(um)","fontsize",4);
ylabel("c/s (cm^-1)","fontsize",4);
title("Normalized Gaussian function V/S distance","fontsize",4);

// Right plot (log scale)
subplot(122);
for i = 1:3
    y = log10((1/(sqrt(%pi)*Dt(i))) * exp(-(x.^2)/(4*(Dt(i)^2))));
    plot(x, y, style=i, rect=[0 -200 4 2]);
end

legend("sqrt(Dt)=0.1","sqrt(Dt)=0.5","sqrt(Dt)=1");
xlabel("Diffusion depth x(um)","fontsize",4);
ylabel("log10(C/S) (cm^-1)","fontsize",4);
title("Normalized Gaussian function V/S distance","fontsize",3);
```

Output: -



Observation: -

- As √(Dt) increases, the concentration profile becomes wider.
- The peak concentration decreases as diffusion increases.
- Dopant spreads deeper into the semiconductor with time.
- The distribution follows a Gaussian curve.
- The logarithmic plot shows exponential decay of concentration with depth.

Application: -

- Used in semiconductor fabrication for controlled doping.
- Formation of p-n junctions.
- Helps in designing diffusion profiles in IC manufacturing.

EXPERIMENT – 4

Aim: - To study the ion implantation process

Apparatus Required: - Scilab Software, Laptop

Theory: - Ion implantation is a modern technique used to introduce dopants into a semiconductor with high precision. In this method, dopant atoms are ionized and accelerated using a high electric field and then directed onto the semiconductor surface.

Unlike diffusion, ion implantation allows accurate control over both **dopant concentration** and **penetration depth**. The implanted ions penetrate the surface and come to rest at an average depth, forming a Gaussian distribution inside the material.

The Physical Principle:

- Dopant atoms are ionized and accelerated.
- These ions strike the semiconductor surface with high energy.
- They penetrate the material and stop at a depth depending on energy.
- The concentration follows a Gaussian distribution around the projected range.

Formula Used: -

$$C(x) = \frac{Q}{\sqrt{2\pi\Delta R_p}} exp(-\frac{(x - R_p)^2}{2\Delta R_p^2})$$

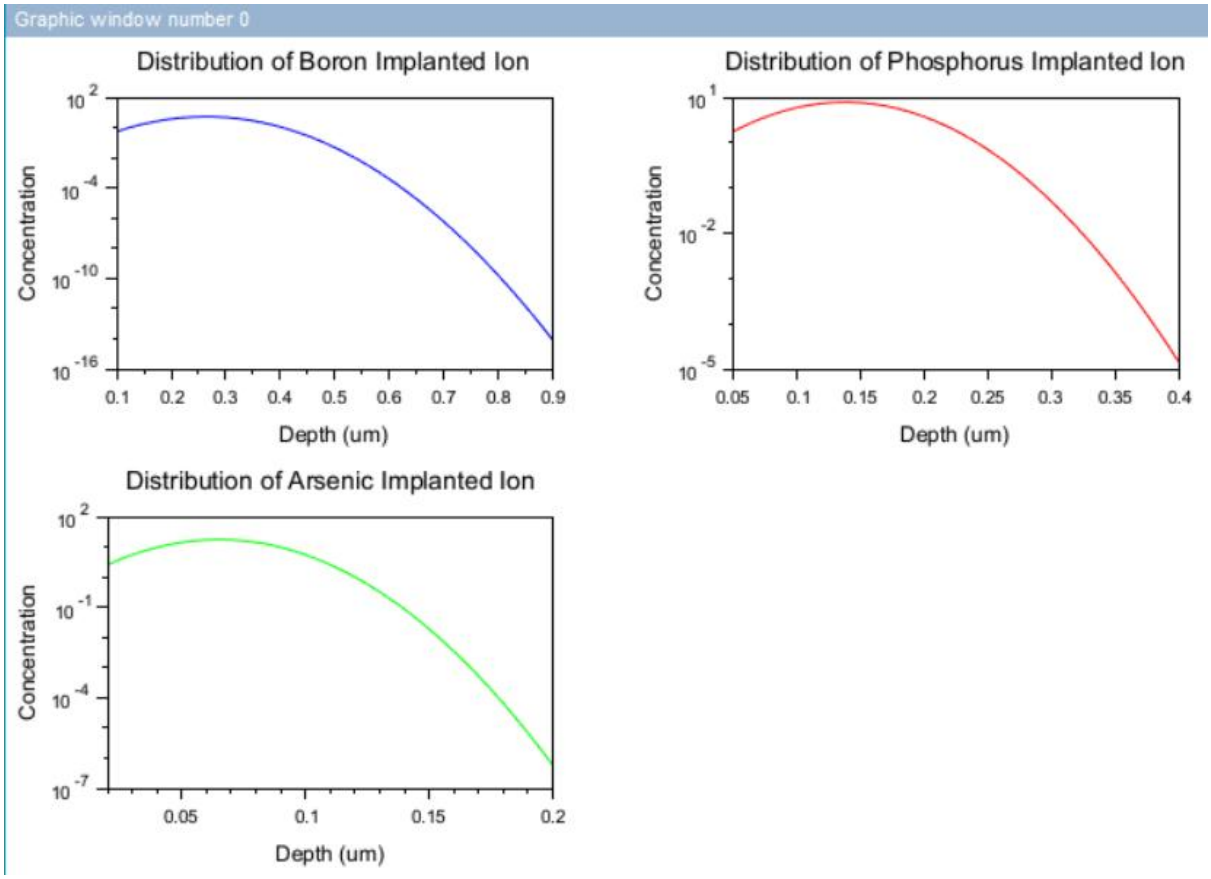
Where:

- $C(x)$: Dopant concentration at depth x
- Q: Implant dose
- R_p: Projected range (average depth)
- ΔR_P: Standard deviation (straggle)
- x: Depth

Code: -

```
clc;
clear;
clf;

x1 = 0.1:0.01:0.9;
Rpb = 0.264;
opb = 0.077;
y1 = (1 ./ (sqrt(2*%pi) * opb)) .* exp(-((x1 - Rpb).^2) ./ (2 * opb^2));

subplot(2,2,1);
plot(x1, y1, 'b');
title("Distribution of Boron Implanted Ion");
xlabel("Depth (um)");
ylabel("Concentration");
gca().log_flags="nl";

x2 = 0.05:0.01:0.4;
Rpp = 0.138;
opp = 0.051;
y2 = (1 ./ (sqrt(2*%pi) * opp)) .* exp(-((x2 - Rpp).^2) ./ (2 * opp^2));

subplot(2,2,2);
plot(x2, y2, 'r');
title("Distribution of Phosphorus Implanted Ion");
xlabel("Depth (um)");
ylabel("Concentration");
gca().log_flags="nl";

x3 = 0.02:0.005:0.2;
RpAs = 0.065;
opAs = 0.023;
y3 = (1 ./ (sqrt(2*%pi) * opAs)) .* exp(-((x3 - RpAs).^2) ./ (2 * opAs^2));

subplot(2,2,3);
plot(x3, y3, 'g');
title("Distribution of Arsenic Implanted Ion");
xlabel("Depth (um)");
ylabel("Concentration");
gca().log_flags="nl";
```

Output: -



Observation: -

- The concentration follows a Gaussian distribution for all dopants.
- Each dopant has a different peak position (projected range).
- Boron penetrates deeper due to its smaller atomic mass.
- Arsenic shows a sharper and narrower distribution due to its heavier mass.
- The log scale clearly shows the rapid decrease in concentration away from the peak.

Application: -

- Used in fabrication of MOSFET source and drain regions.
- Precise doping in integrated circuits.
- Widely used in VLSI and ULSI technology.

EXPERIMENT – 5

Aim: - To study the lithographic process

Apparatus Required: - Scilab Software, Laptop

Theory: - Photolithography is the fundamental pattern transfer technique used in semiconductor manufacturing. It enables the fabrication of micro- and nano-scale structures by selectively exposing a photosensitive material (photoresist) to ultraviolet (UV) light through a patterned mask.

The lithographic process determines the resolution, alignment accuracy, and overall performance of semiconductor devices. Any defects introduced during this stage directly affect the yield, which is the fraction of functional chips produced per wafer.

Yield is strongly dependent on defect density (D_0) and chip area (A). As chip size increases, the probability of defects within a chip also increases, leading to reduced yield. This relationship is commonly modeled using the exponential yield model:

$$Y = e^{-D_0 A}$$

This model assumes random defect distribution across the wafer. Lower defect density and smaller chip size result in higher yield, which is critical for cost-effective IC fabrication.

Types of Photoresists:

- **Positive Photoresist:** Exposed regions become soluble and are removed.
- **Negative Photoresist:** Exposed regions become insoluble and remain.

Steps in Lithographic Process:

1. **Wafer Cleaning:** Removes contaminants and improves adhesion.
2. **Photoresist Coating:** Uniform thin film using spin coating.
3. **Soft Bake:** Removes solvents and stabilizes resist.
4. **Mask Alignment & Exposure:** UV light transfer's pattern.
5. **Development:** Selective removal of resist.
6. **Hard Bake:** Improves durability and adhesion.

Formula Used: -

$$Y = e^{-AD_0}$$

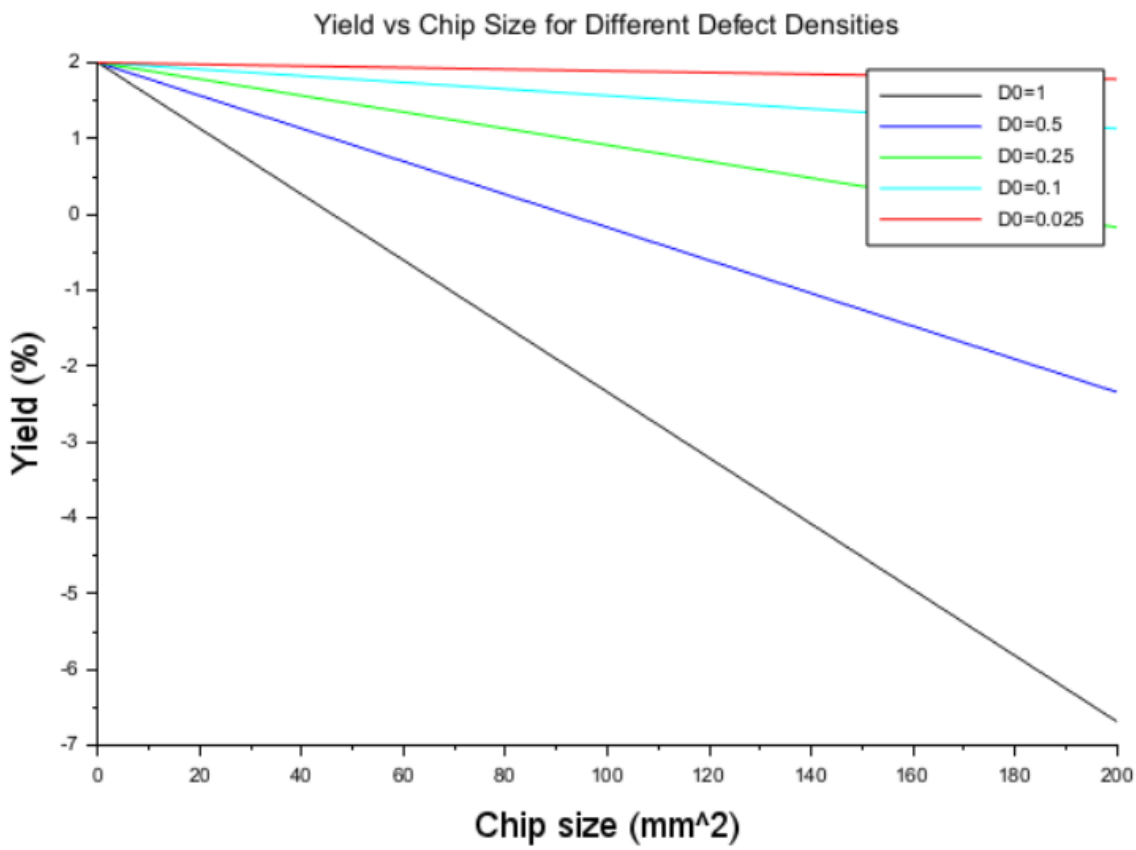
Where:

- Y: Yield (fraction of functional chips)
- D_0: Defect density (defects per unit area)
- A: Chip area

Code: -

```
clc;
clear;
clf;

a = 0:20:200;                    //chip size
D0 = [1 0.5 0.25 0.1 0.025];  //defect density

for i = 1:5
    y = log10((exp(-10*(D0(i)/100)*(a))) * 100);
    plot2d(a, y, style=i);
end

xlabel("Chip size (mm^2)", "fontsize", 4);
ylabel("Yield (%)", "fontsize", 4);
legend("D0=1","D0=0.5","D0=0.25","D0=0.1","D0=0.025");
title("Yield vs Chip Size for Different Defect Densities");
```

Output: -



Observation: -

- Yield decreases exponentially as chip size increases.
- Higher defect density leads to a rapid decrease in yield.
- For low defect density, yield remains high even for larger chip sizes.
- The graph clearly shows that minimizing defects is critical for improving production efficiency.
- The logarithmic scale highlights the exponential nature of yield degradation.

Application: -

- Used in IC fabrication for pattern transfer and design realization.
- Helps in yield estimation and optimization in semiconductor manufacturing.
- Important for cost analysis and scaling in VLSI and ULSI technologies.

EXPERIMENT – 6

Aim: - To study the linear and parabolic rate constants for oxidation (dry and wet) process.

Apparatus Required: - Scilab Software, Laptop

Theory: - Thermal oxidation is a fundamental process in semiconductor fabrication used to grow a silicon dioxide (SiO_2) layer on silicon wafers. This oxide layer serves as an insulator, surface passivation layer, and dielectric in devices such as MOSFETs.

Oxidation of silicon is carried out in two ways:

- **Dry Oxidation:** Uses oxygen (O_2). Produces thin, high-quality oxide layers with slow growth rate.
- **Wet Oxidation**: Uses water vapor (H_2O). Produces thicker oxide layers at a much faster rate but with slightly lower quality.

Oxidation Kinetics:

- **Linear Region (Reaction-controlled):**
  Dominates at initial stages when oxide layer is thin.
  Growth rate is proportional to time:

$$x \propto t$$

Controlled by surface reaction.

- **Parabolic Region (Diffusion-controlled):**
  Dominates at later stages when oxide is thick.
  Growth follows:

$$x \propto \sqrt{t}$$

Controlled by diffusion of oxidizing species through oxide.

Arrhenius Relation:

The temperature dependence of rate constants follows:

$$k = k_0 e^{-E_a/kT}$$

Taking logarithm:

$$\log(k) \propto \frac{-1}{T}$$

Thus, a plot of **log(rate constant) vs 1000/T** gives a straight line.

Formula Used: -

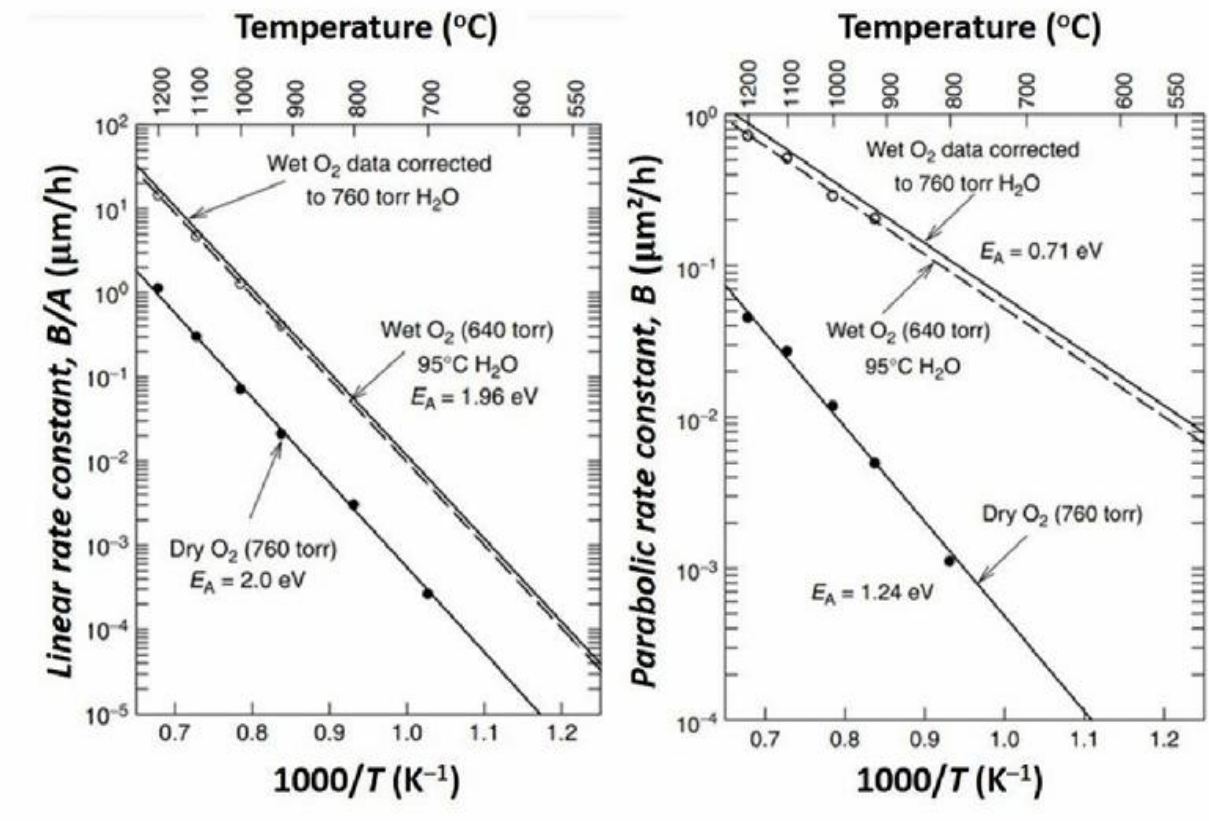
The growth of oxide is described by the Deal–Grove model, which accounts for both reaction and diffusion mechanisms. The model is given by:

$$x^2 + Ax = B(t + \tau)$$

Where:

- x: Oxide thickness
- t: Oxidation time
- A: Linear rate constant
- B: Parabolic rate constant
- τ: Time correction factor

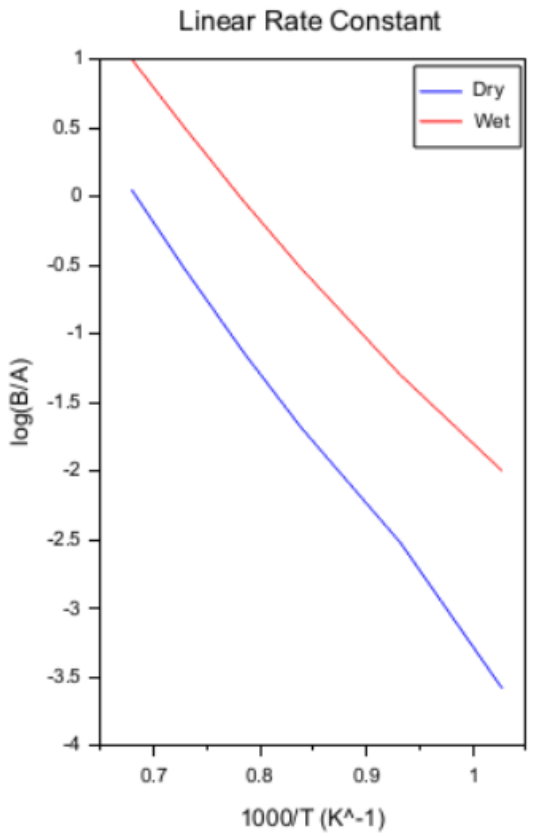
Graph: -

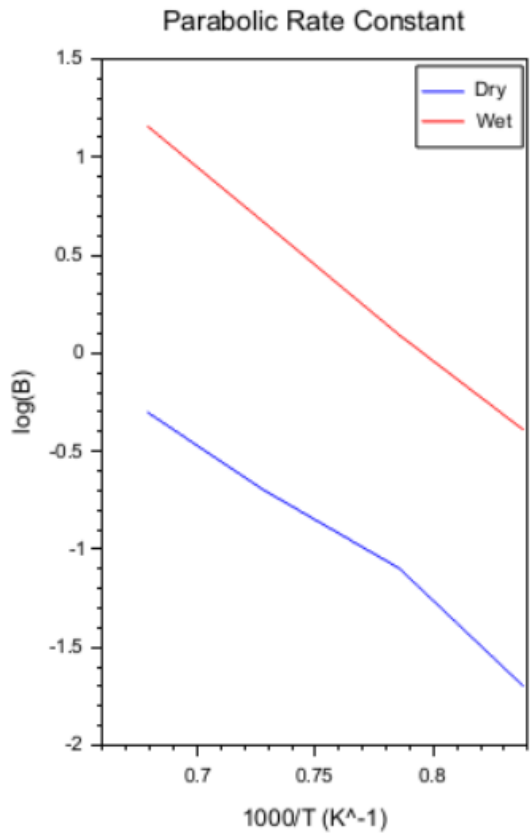


Code: -

```
clc;
clear;
clf;

temp_c = [1200 1100 1000 920 800 700];
T = temp_c + 273.15;
x = 1000 ./ T;

BA_dry = [1.12 0.3 0.071 0.0208 0.003 0.00026];
BA_wet = [10 3.2 0.9 0.3 0.05 0.01];

y_dry = log10(BA_dry);
y_wet = log10(BA_wet);

temp2_c = [1200 1100 1000 920];
T2 = temp2_c + 273.15;
x2 = 1000 ./ T2;

B_dry = [0.5 0.2 0.08 0.02];
B_wet = [14.4 4.64 1.24 0.406];

y2_dry = log10(B_dry);
y2_wet = log10(B_wet);

subplot(121);
plot(x, y_dry, 'b');
plot(x, y_wet, 'r');
legend("Dry","Wet");
xlabel("1000/T (K^-1)");
ylabel("log(B/A)");
title("Linear Rate Constant");

subplot(122);
plot(x2, y2_dry, 'b');
plot(x2, y2_wet, 'r');
legend("Dry","Wet");
xlabel("1000/T (K^-1)");
ylabel("log(B)");
title("Parabolic Rate Constant");
```

Output: -





Observation: -

- Both linear and parabolic rate constants decrease with increasing temperature.
- The plots show straight lines, confirming Arrhenius behaviour.
- Wet oxidation has higher rate constants compared to dry oxidation.
- Linear region dominates at small oxide thickness, while parabolic region dominates at larger thickness.
- The slope of the graph is related to activation energy.

Application: -

- Used in fabrication of gate oxide in MOS devices.
- Formation of insulating layers in IC manufacturing.
- Surface passivation of semiconductor devices.
- Widely used in VLSI and ULSI technology.

EXPERIMENT – 7

AIM - To determine the resistivity of semiconductors by Four probe Method.

APPARATUS REQUIRED – V-Labs

THEORY - At a constant temperature, the resistance, R of a conductor is proportional to its length L and inversely proportional to its area of cross section A.

$$R = \rho \frac{L}{A}$$

A semiconductor has electrical conductivity intermediate in magnitude between that of a conductor and insulator. Semiconductor differs from metals in their characteristic property of decreasing electrical resistivity with increasing temperature.

According to band theory, the energy levels of semiconductors can be grouped into two bands, valence band and the conduction band. In the presence of an external electric field it is electrons in the valence band that can move freely, thereby responsible for the electrical conductivity of semiconductors. In case of intrinsic semiconductors, the Fermi level lies in between the conduction band minimum and valence band maximum. Since conduction band lies above the Fermi level at 0K, when no thermal excitations are available, the conduction band remains unoccupied. So conduction is not possible at 0K, and resistance is infinite. As temperature increases, the occupancy of conduction band goes up, thereby resulting in decrease of electrical resistivity of semiconductor.

Resistivity of semiconductor by four probe method

1. The resistivity of material is uniform in the area of measurement.
2. If there is a minority carrier injection into the semiconductor by the current- carrying electrodes most of the carriers recombine near electrodes so that their effect on conductivity is negligible.
3. The surface on which the probes rest is flat with no surface leakage.
4. The four probes used for resistivity measurement contact surface at points that lie in a straight line.
5. The diameter of the contact between metallic probes and the

semiconductor should be small compared to the distance between the probes.
6. The boundary between the current carrying electrodes and the bulk material is hemispherical and small in diameter.
7. The surface of semiconductor material may be either conducting and non-conducting. A conducting boundary is one on which material of much lower resistivity than semiconductor has been plated. A non-conducting boundary is produced when the surface of the semiconductor is in contact with insulator.

$$\rho = \frac{\rho_0}{f(\frac{w}{s})}$$

Fig: 2 show the resistivity probes on a die of material. If the side boundaries are adequately far from the probes, the die may be considered to be identical to a slice. For this case of a slice of thickness w and the resistivity is computed as

The function, f(w/S) is a divisor for computing resistivity which depends on the value of w and S
We assume that the size of the metal tip is infinitesimal and sample thickness is greater than the distance between the probes,

$$\rho_0 = \frac{V}{I} \times 2\pi S$$

Temperature dependence of resistivity of semiconductor

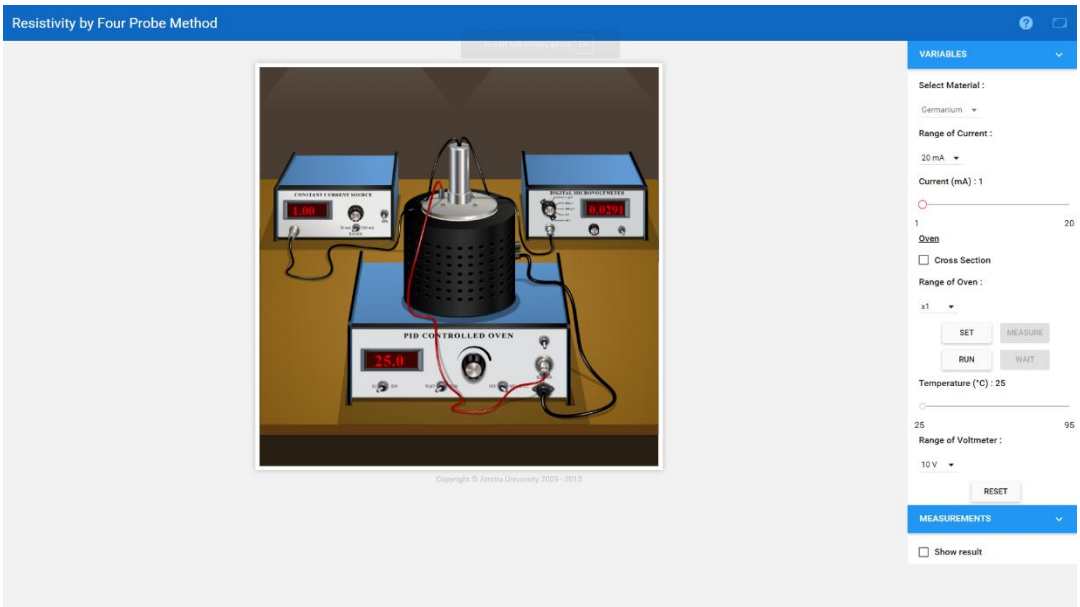
Total electrical conductivity of a semiconductor is the sum of the conductivities of the valence band and conduction band carriers. Resistivity is the reciprocal of conductivity and its temperature dependence is given by

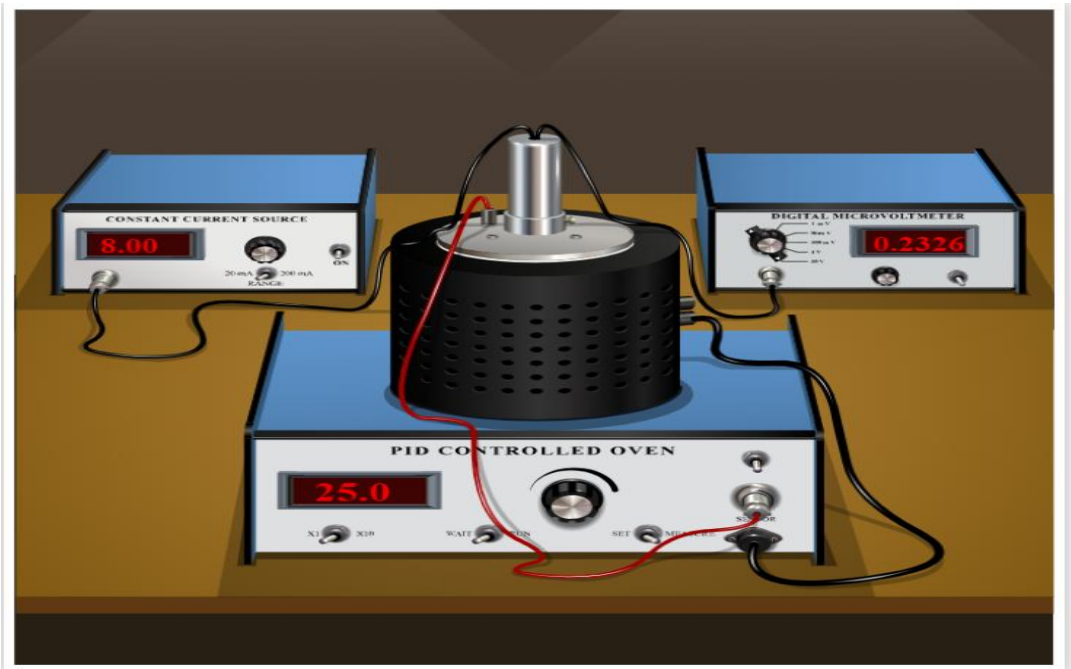
$$\rho = A \exp \frac{E_g}{2KT}$$

Applications

1. Remote sensing areas
2. Resistance thermometers
3. Induction hardening process
4. Accurate geometry factor estimation
5. Characterization of fuel cells bipolar plates

<u>CIRCUIT DIAGRAM</u> -





Observation Table :

For germanium at 8 mA

S.No.	Temperature	Voltage(V)	Resistivity (ohm-cm)
1	30	0.2257	6.0172
2	35	0.2193	5.8444
3	40	0.2132	5.6819
4	45	0.2074	5.5288
5	50	0.2020	5.3843

For silicon at 10 mA

S.No.	Temperature	Voltage(V)	Resistivity (ohm-cm)
1	30	1.1986	25.5598
2	35	1.1372	24.2497
3	40	1.0807	23.0456
4	45	1.0827	21.9363
5	50	0.9807	20.9123